```
root@testPSQS01:/backup/postgres/pg_log# apt install -y python3-pip python3-dev libpq-dev build-essential
Чтение списков пакетов… Готово
Построение дерева зависимостей… Готово
Чтение информации о состоянии… Готово
Будут установлены следующие дополнительные пакеты:
  cpp cpp-11 dpkg-dev fakeroot fontconfig-config fonts-dejavu-core g++ g++-11 gcc gcc-11 gcc-11-base javas
  libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libatomic1 libc-dev-bi
  libcc1-0 libcrypt-dev libdeflate0 libexpat1-dev libfakeroot libfontconfig1 libgcc-11-dev libgd3 libgomp1
```

```
(airflow_venv) root@testPSQS01:/# pip install apache-airflow
Collecting apache-airflow
  Downloading apache_airflow-3.1.8-py3-none-any.whl (12 kB)
Collecting apache-airflow-core==3.1.8
  Downloading apache_airflow_core-3.1.8-py3-none-any.whl (3.9 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.9/3.9 MB 790.0 kB/s eta 0:00:00
Collecting apache-airflow-task-sdk==1.1.8
  Downloading apache_airflow_task_sdk-1.1.8-py3-none-any.whl (305 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 305.7/305.7 KB 532.6 kB/s eta 0:00:00
Collecting svcs>=25.1.0
  Downloading svcs-25.1.0-py3-none-any.whl (19 kB)
Collecting asgiref>=2.3.0
  Downloading asgiref-3.11.1-py3-none-any.whl (24 kB)
Collecting pydantic>=2.11.0
  Downloading pydantic-2.12.5-py3-none-any.whl (463 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 463.6/463.6 KB 672.5 kB/s eta 0:00:00
```

```
bove.
(airflow_venv) root@testPSQS01:/var# airflow api-server --port 8080
2026-04-04T11:56:16.262644Z [info     ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-04-04T11:56:16.262852Z [info     ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-04-04T11:56:16.262941Z [info     ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-04-04T11:56:16.263017Z [info     ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:
2026-04-04T11:56:16.263091Z [info     ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-04-04T11:56:16.263179Z [info     ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
  ____________       _____________
 ____    |__( )_________  __/__  /________      __
____  /| |_  /__  ___/_  /_ __  /_  __ \_ | /| / /
___  ___ |  / _  /   _  __/ _  / / /_/ /_ |/ |/ /
 _/_/  |_/_/  /_/    /_/    /_/  \____/____/|__/
2026-04-04T11:56:17.244653Z [info     ] Running the uvicorn with:
Apps: all
Workers: 1
Host: 0.0.0.0:8080
Timeout: 120
Logfiles: -
================================================================ [airflow.cli.commands.api_server_command] loc=api_server_comm
/airflow_venv/lib/python3.10/site-packages/airflow/api_fastapi/execution_api/routes/__init__.py:23 DeprecationWarning: 'HTTP_42
BLE_ENTITY' is deprecated. Use 'HTTP_422_UNPROCESSABLE_CONTENT' instead.
2026-04-04T11:56:17.671913Z [warning  ] Could not import pandas. Holidays will not be considered. [airflow.example_dags.plugins
c=workday.py:41
Simple auth manager | Password for user 'admin': wVDCFVP8q4Q5rqtG
```

```
CREATE TABLE IF NOT EXISTS currency_rates
(
    `parse_date` Date DEFAULT today() COMMENT 'Дата парсинга',
    `date_rates` DateTime COMMENT 'Дата, на которую установлены курсы (поле Date из JSON)',
    `previous_date_rates` DateTime COMMENT 'Предыдущая дата курсов',
    `timestamp_rates` DateTime COMMENT 'Временная метка публикации курсов',
    `currency_id` String COMMENT 'Внутренний идентификатор валюты (ID)',
    `num_code` String COMMENT 'Цифровой код валюты (NumCode)',
    `char_code` String COMMENT 'Буквенный код валюты (CharCode)',
    `nominal` UInt32 COMMENT 'Номинал (Nominal)',
    `currency_name` String COMMENT 'Название валюты (Name)',
    `value` Float64 COMMENT 'Курс (Value)',
    `previous_value` Float64 COMMENT 'Предыдущий курс (Previous)'
)
ENGINE = MergeTree
ORDER BY (parse_date, char_code)
COMMENT 'Таблица для хранения курсов валют с сайта ЦБ РФ (ежедневный парсинг)'

Query id: 3d0e29c8-d6b3-4491-8fee-86a349a1d6b3

Ok.

0 rows in set. Elapsed: 0.013 sec.
```

Не защищено 10.1.13.79:8080

Welcome

Stats

0 Failed Dags

0 Running Dags

0 Active Dags

First 10 favorite Dags

No favorites yet. Click the star icon next to a Dag in the list to add it to your favorites.

Health

MetaDatabase

Scheduler

Triggerer

Dag Processor

Pool Slots

History

Last 24 Hours

2026-04-03 15:16:50 - 2026-04-04 15:16:50

0 Dag Runs

0 Queued

0 Running

0 Success

0 Failed

```
/root/airflow/dags/currency_exchange_rate.py   [----] 66
    url = 'https://www.cbr-xml-daily.ru/daily_json.js'
    resp = requests.get(url)
    resp.raise_for_status()
    data = resp.json()

    # 2. Извлечь общие даты
    date_rates = data['Date']
    previous_date = data['PreviousDate']
    timestamp = data['Timestamp']

    # 3. Подготовить строки для вставки
    rows = []
    parse_date = datetime.now().date()   # дата парсинга

    for code, val in data['Valute'].items():
        rows.append({
            'parse_date': parse_date,
            'date_rates': date_rates,
            'previous_date_rates': previous_date,
            'timestamp_rates': timestamp,
            'currency_id': val.get('ID'),
            'num_code': val.get('NumCode'),
            'char_code': val.get('CharCode'),
            'nominal': val.get('Nominal'),
            'currency_name': val.get('Name'),
            'value': val.get('Value'),
            'previous_value': val.get('Previous'),
        })

    # 4. Подключение к ClickHouse и вставка данных
    client = Client(**CLICKHOUSE_CONFIG)
    insert_query = '''
        INSERT INTO currency_rates (
```

Dag
currency_exchange_rate

Dag Run
manual__2026-04-08T13:10:12.459005+00:00

Options

00:05:07
00:02:33

get_rates

manual__2026-04-08T13:10:12.459005+00:00 Success

Logical Date	Run Type	Start Date	End Date
2026-04-08 16:10:05	manual	2026-04-08 16:10:13	2026-04-08

Task Instances | Asset Events | Audit Log | Code | Details

Search Tasks

All States

1 Task Instance

Task ID	Map Index	State	Start Date
get_rates		Success	2026-04-08 16:10:13

```
SELECT *
FROM currency_rates
LIMIT 15

Query id: 41f75a0d-4710-4e71-8ee8-05dc759cea09

    ┌─parse_date─┬──────────date_rates─┬─previous_date_rates─┬──────timestamp_rates─┬─currency_id─┬─num_code─┬─char_code─┬─nominal─┬
cy_name──────────┬─────value─┬─previous_value─┐
 1. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01230      │ 784      │ AED       │       1 │
 ОАЭ             │    21.443 │        21.4371 │
 2. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01060      │ 051      │ AMD       │     100 │
ких драмов       │   20.9179 │        20.9037 │
 3. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01010      │ 036      │ AUD       │       1 │
лийский доллар   │   54.4632 │        54.1568 │
 4. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01020A     │ 944      │ AZN       │       1 │
йджанский манат  │   46.3233 │        46.3104 │
 5. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01685      │ 050      │ BDT       │     100 │
                 │   64.1074 │        64.0392 │
 6. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01080      │ 048      │ BHD       │       1 │
нский динар      │  209.3952 │        209.337 │
 7. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01105      │ 068      │ BOB       │       1 │
ано              │   11.3965 │        11.3933 │
 8. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01115      │ 986      │ BRL       │       1 │
ьский реал       │   15.2826 │        15.2419 │
 9. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01090B     │ 933      │ BYN       │       1 │
сский рубль      │   26.9755 │        26.9237 │
10. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01350      │ 124      │ CAD       │       1 │
кий доллар       │   56.5892 │        56.5654 │
11. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01775      │ 756      │ CHF       │       1 │
рский франк      │    98.659 │        98.4589 │
12. │ 2026-04-08 │ 2026-04-08 08:30:00 │ 2026-04-07 08:30:00 │ 2026-04-08 13:00:00 │ R01375      │ 156      │ CNY       │       1 │
```